Problem Set 5

9.14 ⋆⋆ I am spinning a bucket of water about its vertical axis with angular velocity Ω. Show that, once the water has settled in equilibrium (relative to the bucket), its surface will be a parabola. (Use cylindrical polar coordinates and remember that the surface is an equipotential under the combined effects of the gravitational and centrifugal forces.)

9.15 ⋆⋆ On a certain planet, which is perfectly spherically symmetric, the free-fall acceleration has magnitude $g = g_0$ at the North Pole and $g = \lambda g_0$ at the equator (with $0 \leq \lambda \leq 1$). Find $g(\theta)$, the free-fall acceleration at colatitude θ as a function of θ.

9.9 ⋆ A bullet of mass m is fired with muzzle speed v_0 horizontally and due north from a position at colatitude θ. Find the direction and magnitude of the Coriolis force in terms of m, v_0, θ, and the earth's angular velocity Ω. How does the Coriolis force compare with the bullet's weight if $v_0 = 1000$ m/s and $\theta = 40$ deg?

9.10 ⋆⋆ The derivation of the equation of motion (9.34) for a rotating frame made the assumption that the angular velocity $\boldsymbol{\Omega}$ was constant. Show that if $\dot{\boldsymbol{\Omega}} \neq 0$ then there is a third "fictitious force," sometimes called the *azimuthal force*, on the right side of (9.34) equal to $m\mathbf{r} \times \dot{\boldsymbol{\Omega}}$.

9.20 ⋆⋆ Consider a frictionless puck on a horizontal turntable that is rotating counterclockwise with angular velocity Ω. **(a)** Write down Newton's second law for the coordinates x and y of the puck as seen by me standing on the turntable. (Be sure to include the centrifugal and Coriolis forces, but ignore the earth's rotation.) **(b)** Solve the two equations by the trick of writing $\eta = x + iy$ and guessing a solution of the form $\eta = e^{-i\alpha t}$. [In this case — as in the case of critically damped SHM discussed in Section 5.4 — you get only one solution this way. The other has the same form (5.43) we found for the second solution in damped SHM.] Write down the general solution. **(c)** At time $t = 0$, I push the puck from position $\mathbf{r}_0 = (x_0, 0)$ with velocity $\mathbf{v}_0 = (v_{x0}, v_{y0})$ (all as measured by me on the turntable). Show that

$$\left.\begin{aligned} x(t) &= (x_0 + v_{x0}t)\cos\Omega t + (v_{y0} + \Omega x_0)t\sin\Omega t \\ y(t) &= -(x_0 + v_{x0}t)\sin\Omega t + (v_{y0} + \Omega x_0)t\cos\Omega t \end{aligned}\right\}. \tag{9.72}$$

(d) Describe and sketch the behavior of the puck for large values of t. [*Hint:* When t is large the terms proportional to t dominate (except in the case that both their coefficients are zero). With t large, write (9.72) in the form $x(t) = t(B_1\cos\Omega t + B_2\sin\Omega t)$, with a similar expression for $y(t)$, and use the trick of (5.11) to combine the sine and cosine into a single cosine — or sine, in the case of $y(t)$. By now you can recognize that the path is the same kind of spiral, whatever the initial conditions (with the one exception mentioned).]

9.22 ⋆⋆ If a negative charge $-q$ (an electron, for example) in an elliptical orbit around a fixed positive charge Q is subjected to a weak uniform magnetic field $\mathbf{B}$, the effect of $\mathbf{B}$ is to make the ellipse precess slowly — an effect known as **Larmor precession**. To prove this, write down the equation of motion of the negative charge in the field of Q and $\mathbf{B}$. Now rewrite it for a frame rotating with angular velocity $\boldsymbol{\Omega}$. [Remember that this changes both $d^2\mathbf{r}/dt^2$ and $d\mathbf{r}/dt$.] Show that by suitable choice of $\boldsymbol{\Omega}$ you can arrange that the terms involving $\dot{\mathbf{r}}$ cancel out, but that you are left with one term involving $\mathbf{B} \times (\mathbf{B} \times \mathbf{r})$. If $\mathbf{B}$ is weak enough this term can certainly be neglected. Show that in this case the orbit in the rotating frame is an ellipse (or hyperbola). Describe the appearance of the ellipse as seen in the original nonrotating frame.

9.23 ⋆⋆ Here is an unusual way to solve the two-dimensional isotropic oscillator — the motion of a particle subject to a force $-k\mathbf{r}$. Show that by choosing a suitable rotating reference frame, you can arrange that the centrifugal force exactly cancels the force $-k\mathbf{r}$. Recalling the analogy between the Coriolis and magnetic forces, you should be able to write down the general solution for the motion as seen in the rotating frame. If you write your solution in the complex form of Section 2.7, then you can transform back to the nonrotating frame by multiplying by a suitable rotating complex number. Show that the general solution is an ellipse. [See Problem 8.11 for some guidance on this last part.]

9.26 ⋆⋆ In Section 9.8, we used a method of successive aproximations to find the orbit of an object that is dropped from rest, correct to first order in the earth's angular velocity Ω. Show in the same way that if an object is thrown with initial velocity $\mathbf{v}_0$ from a point O on the earth's surface at colatitude θ, then to first order in Ω its orbit is

$$\left.\begin{aligned} x &= v_{xo}t + \Omega(v_{yo}\cos\theta - v_{zo}\sin\theta)t^2 + \tfrac{1}{3}\Omega g t^3 \sin\theta \\ y &= v_{yo}t - \Omega(v_{xo}\cos\theta)t^2 \\ z &= v_{zo}t - \tfrac{1}{2}gt^2 + \Omega(v_{xo}\sin\theta)t^2. \end{aligned}\right\} \tag{9.73}$$

[First solve the equations of motion (9.53) in zeroth order, that is, ignoring Ω entirely. Substitute your zeroth-order solution for $\dot{x}$, $\dot{y}$, and $\dot{z}$ into the right side of equations (9.53) and integrate to give the next approximation. Assume that v_0 is small enough that air resistance is negligible and that $\mathbf{g}$ is a constant throughout the flight.]

9.30 ⋆⋆⋆ The Coriolis force can produce a torque on a spinning object. To illustrate this, consider a horizontal hoop of mass m and radius r spinning with angular velocity ω about its vertical axis at colatitude θ. Show that the Coriolis force due to the earth's rotation produces a torque of magnitude $m\omega\Omega r^2 \sin\theta$ directed to the west, where Ω is the earth's angular velocity. This torque is the basis of the gyrocompass.

9.34 ⋆⋆⋆ At a point P on the earth's surface, an enormous perfectly flat and frictionless platform is built. The platform is exactly horizontal — that is, perpendicular to the local free-fall acceleration $\mathbf{g}_P$. Find the equation of motion for a puck sliding on the platform and show that it has the same form as (9.61) for the Foucault pendulum except that the pendulum's length L is replaced by the earth's radius R. What is the frequency of the puck's oscillations and what is that of its Foucault precession? [*Hints:* Write the puck's position vector, relative to the earth's center O as $\mathbf{R} + \mathbf{r}$, where $\mathbf{R}$ is the position of the point P and $\mathbf{r} = (x, y, 0)$ is the puck's position relative to P. The contribution to the centrifugal force involving $\mathbf{R}$ can be absorbed into $\mathbf{g}_P$ and the contribution involving $\mathbf{r}$ is negligible. The restoring force comes from the variation of $\mathbf{g}$ as the puck moves.] To check the validity of your approximations, compare the approximate size of the gravitational restoring force, the Coriolis force, and the neglected term $m(\boldsymbol{\Omega} \times \mathbf{r}) \times \boldsymbol{\Omega}$ in the centrifugal force.